TECHNICAL ANALYTICS REPORT

From Rukpokwu/Sars Road to UNIPORT: A Case Study Analysis of Bolt Fare Pricing and Driver Earnings in Port Harcourt

An investigation into why Bolt drivers in Port Harcourt charge above the app's quoted fare, and why Bolt has not revised its in-app pricing for the Nigerian market

Prepared by	Ibialabo Dappa
Programmed By	Harvoxx Tech Hub — Data Analytics Accelerator Program
Case Study Route	Rukpokwu/SARS Road → University of Port Harcourt (UNIPORT)
Location	Port Harcourt, Rivers State, Nigeria
Sample Size	116 real Bolt trip observations
Tools Used	Microsoft Excel (Data Analysis ToolPak, PivotTables, Regression, Correlation)
Report Type	Data Analytics Case Study & Dashboard Report

Table of Contents

1. Executive Summary **3**

2. Introduction and Problem Statement **3**

3. Objectives of the Study **4**

4. Data Description **4**

5. Data Cleaning and Preparation **5**

6. Analytical Approach **5**

7. Results and Visualisations **6**

Insight 1: Highest-Revenue Time of Day 6

Insight 2: Highest-Revenue Vehicle 6

Insight 3: Bolt Fare vs. Fuel Cost by Time Range 7

Insight 4: Number of Rides vs. Bolt Fare 8

Insight 5: Vehicle Choice vs. Fare and Fuel Cost 8

Insight 6: Why Bolt Has Not Updated Its Fare 9

Interactive Dashboard 10

8. Interpretation and Business Impact **10**

9. Recommendations **10**

10. Limitations and Next Steps **11**

10.1 Fuel Cost as an Estimated Proxy Variable 11

10.2 Fuel Cost and Driver Take-Home Earnings — A Nigerian Perspective 11

10.3 Sample Size and Scope 12

11. Conclusion **12**

References **13**

1. Executive Summary

This project examines why Bolt drivers in Port Harcourt frequently charge fares above the amount quoted on the Bolt app, and why Bolt has not adjusted its in-app pricing to reflect the rising and fluctuating cost of fuel in Nigeria. The Rukpokwu–SARS Road–UNIPORT route was used as a representative case study route, and an interactive Excel dashboard was built to support the analysis.

The study is based on 116 real Bolt trip records, each capturing the time of the ride, pickup location, destination, time-of-day range, vehicle plate number, vehicle type, distance travelled, the fare quoted by the Bolt app, the fare actually negotiated by the driver, the number of completed rides per driver, the fuel price per litre at the time of the ride, the driver's take-home pay after Bolt's 25% commission, and the fuel efficiency range (in kilometres per litre) for each vehicle. For every vehicle, a low (worst-case) and high (best-case) fuel efficiency figure was used to estimate a low and high fuel cost; the low, worst-case fuel cost estimate was used throughout the analysis to represent a conservative, upper-bound view of what driving actually costs. Excel was used both to clean the dataset and to carry out the exploratory analysis, including PivotTables, correlation analysis, and regression modelling.

The results show that a driver's estimated fuel cost explains only about 25.8% ($R^2 = 0.258$) of the variation in what a driver actually takes home per ride. In other words, fuel cost alone does not strongly predict driver earnings; other factors, such as distance and time of day, matter more. This weak relationship may help explain Bolt's reluctance to revise its fare structure: if Bolt's own pricing model, like the one used in this study, relies on generalised or manufacturer-stated fuel efficiency figures rather than the real day-to-day cost pressures drivers face — including fuel price fluctuation, government policy, and heavy traffic — the platform may simply be underestimating the true cost of operating in a market like Nigeria.

The analysis further shows that Bolt generates the most revenue for drivers during the evening period, from 5:00 p.m. onward, followed by the early-morning rush between 7:00 a.m. and 9:00 a.m. The Toyota Corolla and Toyota Camry were the most frequently used vehicles among the drivers sampled, largely because of their relatively low fuel consumption, affordable maintenance, and the wide availability of spare parts in Nigeria. The data also shows that the number of rides a driver has completed has no bearing on the fare charged by the Bolt app — a driver with a thousand completed rides is quoted the same fare as a first-time driver for the same trip.

Based on these findings, the report recommends that Bolt introduce a structured driver feedback system covering fuel price and traffic conditions (not only passenger star ratings), review and update its fare algorithm at more frequent intervals — particularly in markets where fuel regulation has changed, such as Nigeria following the removal of the fuel subsidy — and consider a loyalty or reward system for drivers with a strong ride history, to improve retention on both the driver and passenger sides of the platform.

2. Introduction and Problem Statement

Ride-hailing platforms such as Bolt have become a major part of urban transport in Nigerian cities, including Port Harcourt. Bolt sets a fare for every trip through its app pricing algorithm, but in practice, many drivers do not accept this quoted fare. Instead, they negotiate a higher amount directly with the passenger before or during the ride. This pattern has become common enough that it raises an important question: what is driving the gap between the price Bolt quotes and the price drivers actually charge?

This project set out to investigate that gap using real trip data. It focuses on three connected questions. First, why do drivers charge more than the Bolt app's quoted fare? Second, does the cost of fuel — which has risen sharply in Nigeria following the removal of the fuel subsidy — meaningfully explain how much a driver takes home per ride? Third, why has Bolt not adjusted its fare structure in response to these rising costs, despite visible pressure from drivers?

To ground the analysis in a real, relatable scenario, the study uses the Rukpokwu/Sars road to UNIPORT route as an anchoring case study. This is a common route for Bolt drivers operating in Port Harcourt, and it reflects the mix of residential, commercial, and traffic-heavy conditions typical of the wider dataset. The findings from this route, and from the broader sample of 116 trips it forms part of, are intended to offer a realistic, data-backed perspective on driver earnings and fare pricing that is relevant not only to Port Harcourt, but to the broader Nigerian ride-hailing market.

3. Objectives of the Study

- To determine why Bolt drivers in Port Harcourt frequently charge fares above the amount quoted on the app.
- To assess whether estimated fuel cost meaningfully explains driver take-home earnings.
- To identify which time of day and which vehicle types generate the highest revenue on the Bolt platform.
- To test whether the number of rides a driver has completed has any effect on the fare quoted by the Bolt app.
- To build an interactive Excel dashboard that presents these findings clearly for both technical and non-technical audiences.
- To propose practical recommendations that Bolt and similar ride-hailing platforms could consider in response to the findings.

4. Data Description

The dataset consists of 116 real Bolt trip records collected directly from ride activity in Port Harcourt. Each record represents a single completed trip and includes the following categories of information:

- Trip details: date and time of the ride, pickup location, destination, and time-of-day range (7:00 a.m.–9:00 a.m., 12:00 p.m.–2:00 p.m., 5:00 p.m.–9:00 p.m., and 9:00 p.m. onward).
- Vehicle details: car plate number, vehicle model, and distance travelled per trip.
- Fare details: the fare quoted by the Bolt app, the fare actually negotiated by the driver, and the driver's take-home amount after Bolt's 25% commission.
- Driver activity: the number of rides completed by each driver at the time of the trip.
- Fuel details: the fuel price per litre on the day of the ride and each vehicle's fuel efficiency range in kilometres per litre, expressed as a low (worst-case) and high (best-case) figure.
- Derived fuel cost estimates: a low estimated fuel cost (based on worst-case efficiency) and a high estimated fuel cost (based on best-case efficiency) for each trip. The low, worst-case estimate was used throughout the analysis, since it offers a conservative, upper-bound view of what a trip is likely to cost the driver in fuel alone.

The time frame of the study spans four time-of-day bands — 7:00 a.m. to 9:00 a.m., 12:00 p.m. to 2:00 p.m., 5:00 p.m. to 9:00 p.m., and 9:00 p.m. onward — and the case study route (Rukpokwu/Sars Road to UNIPORT) was used to anchor the analysis in real, recognisable trip patterns in Port Harcourt.

5. Data Cleaning and Preparation

The raw data was collected directly from real Bolt trips and transferred into a CSV file for cleaning and analysis in Excel. The following steps were carried out to prepare the data:

- Outlier treatment: distance values were cross-checked and corrected using Google Maps, since a small number of recorded distances appeared inconsistent with the actual route — likely due to manual entry.
- Feature engineering: several new variables were created to support the analysis, including the driver's take-home amount after Bolt's 25% commission, each vehicle's fuel efficiency range, the estimated fuel used for each vehicle using the worst and normal scenario (low & high), the estimated low and high fuel cost per trip.
- Categorical handling: fields such as vehicle type and time-of-day range were standardised and grouped consistently to allow for reliable comparison across the dataset.

These steps ensured that the dataset used for analysis was consistent, free of obvious data-entry errors, and structured in a way that made the subsequent regression and correlation analysis meaningful.

6. Analytical Approach

The analysis was carried out entirely in Microsoft Excel, using a combination of tools suited to exploratory and statistical analysis:

- PivotTables and PivotCharts were used to summarise revenue by time-of-day range and by vehicle type.
- The Data Analysis ToolPak was used to run correlation analysis and multiple linear regression in order to test the relationship between fare, distance, fuel cost, and driver take-home earnings.
- Scatter plots with trendlines were used to visually assess the relationship between key variable pairs, such as negotiated fare against fuel cost, and driver take-home amount against estimated fuel cost.
- Conditional formatting (colour scales) was used to build a correlation heatmap, making it easier to spot which variables moved together for testing.
- An interactive dashboard was built using slicers connected to PivotTables, allowing the findings to be filtered dynamically by time range, vehicle type, and the car efficiency range (Km/L).

Regression output was interpreted using three key indicators: the R Square value, which shows how much of the variation in the outcome is explained by the model; the p-value for each variable, which shows whether that variable's effect is statistically meaningful; and the coefficient, which shows the size and direction of each variable's effect.

7. Results and Visualisations

This section presents the six key insights generated from the analysis, followed by the full interactive dashboard.

Insight 1: Which Time of Day Generates the Highest Revenue on the Bolt App

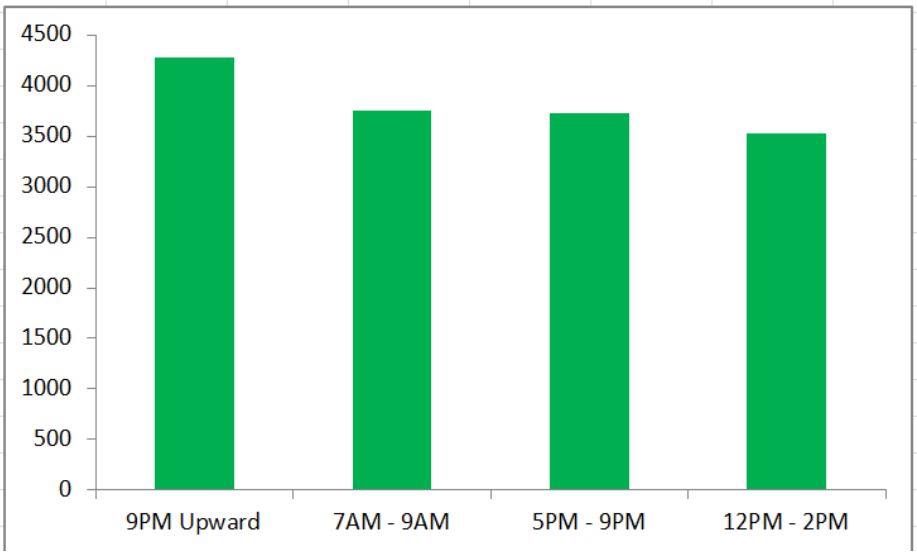


Figure 1. Average or total fare revenue by time-of-day range.

Revenue on the Bolt platform is highest during the evening period, from 5:00 p.m. onward, followed by the early-morning rush between 7:00 a.m. and 9:00 a.m. These two windows correspond to peak commuting periods in Port

Harcourt, when demand for rides is at its highest and drivers are able to complete more trips or command better fares.

Insight 2: Which Vehicle Generates the Highest Revenue on the Bolt App

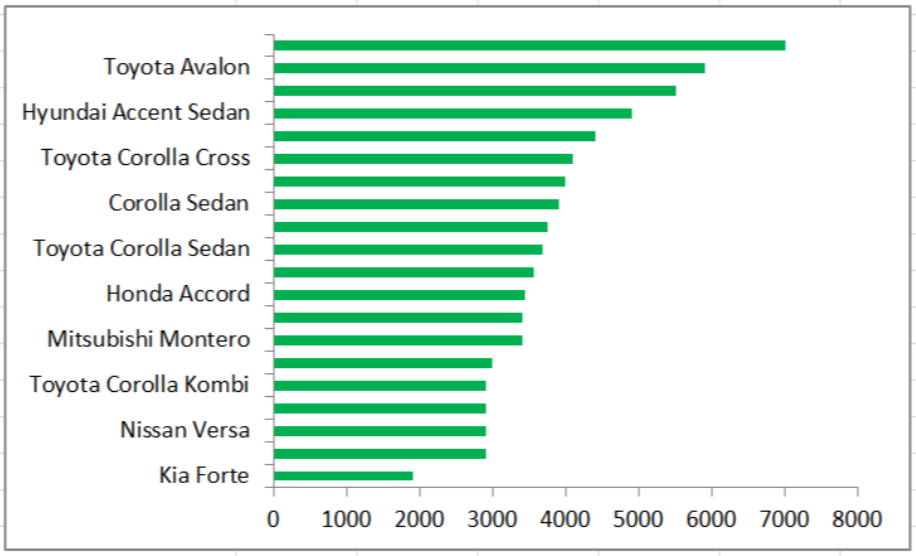


Figure 2. Total or average revenue generated by vehicle model.

The Toyota Corolla and Toyota Camry generate the most revenue among the vehicles sampled. This is notable given that Bolt's fare structure is not vehicle specific; the platform does not assign a fixed or premium fare based on the type or class of car used. The elevated revenue associated with these models is therefore more plausibly attributable to their prevalence among Bolt drivers in Nigeria, rather than any inherent fare advantage. This popularity is likely driven by relatively low fuel consumption, affordable and widely available spare parts, and manageable maintenance costs compared to non-Toyota alternatives. It is worth noting that if Bolt's pricing model did account for vehicle class or perceived value, a higher-end vehicle such as the Mercedes-Benz would reasonably be expected to generate the highest revenue instead.

Insight 3: Comparison of Bolt Fare and Fuel Cost per Distance, by Time Range

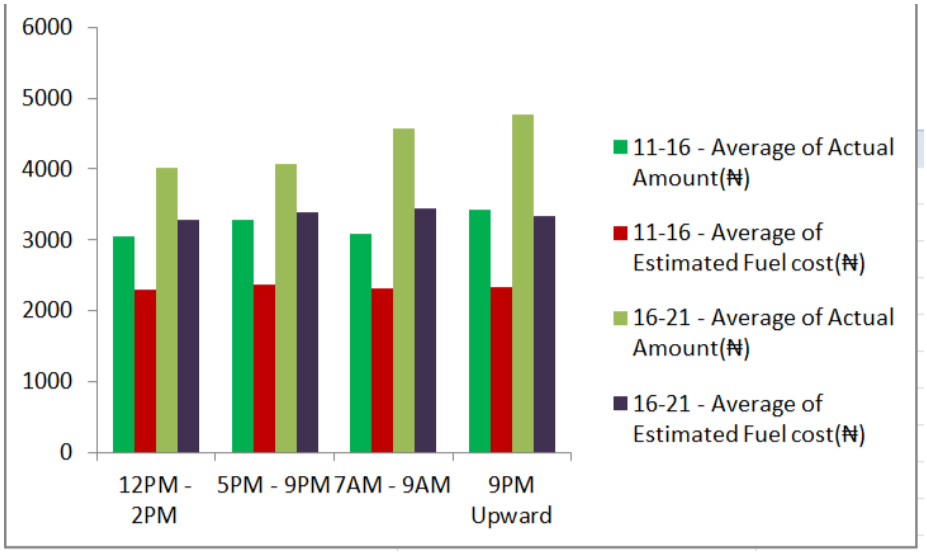


Figure 3. Bolt fare and estimated fuel cost per trip, compared across time-of-day ranges.

This chart compares the Bolt app's quoted fare and the estimated fuel cost against the distance travelled, across different times of day. The gap between fare and fuel cost is not constant — it narrows or widens depending on when the trip takes place and how far it covers. This shows that Bolt's one-size-fits-all pricing doesn't adjust for how conditions, like distance, traffic, and demand, change throughout the day.

Insight 4: Does the Number of Rides Completed by a Driver Affect the Bolt Fare?

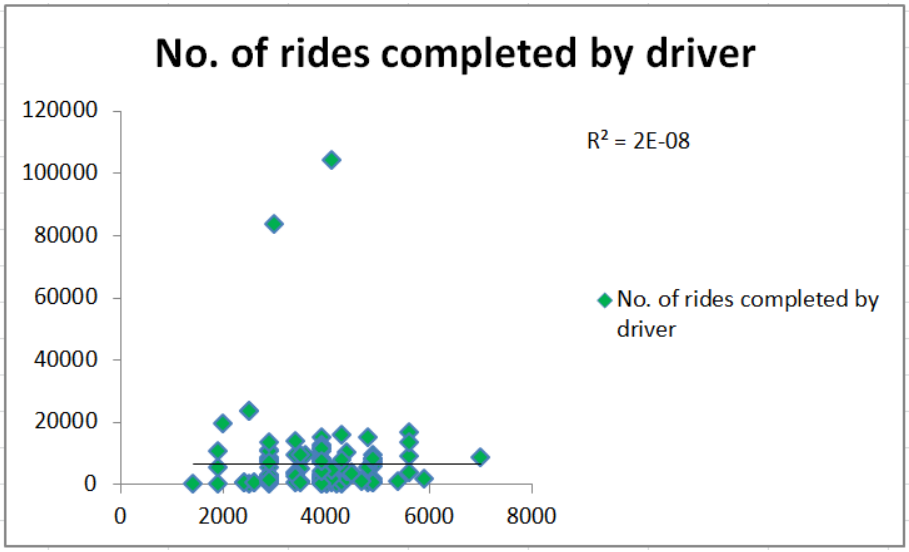


Figure 4. Number of rides completed by driver, plotted against Bolt app fare.

The data shows no meaningful relationship between the number of rides a driver has completed and the fare quoted by the Bolt app. A driver with a thousand completed rides is quoted the same fare as a driver completing their very first ride, for the same trip. This confirms that Bolt's pricing algorithm does not currently reward driver experience or loyalty in any way.

Insight 5: Why Drivers Use Specific Vehicles, Compared Against Bolt Fare and Fuel Cost

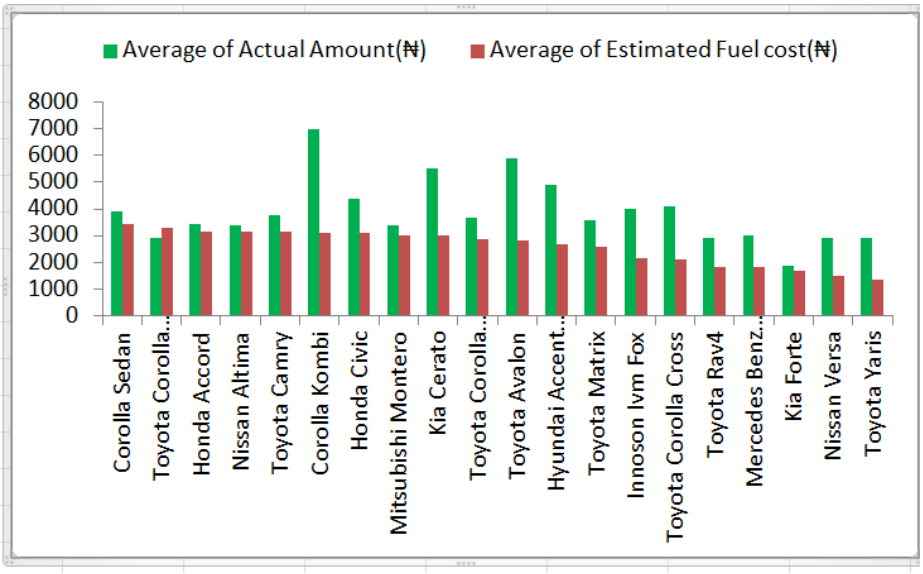


Figure 5. Vehicle type compared against Bolt fare and estimated fuel cost.

Drivers appear to favour vehicles that offer the best balance between fuel efficiency and the fare they can expect to earn. This explains the strong preference for the Toyota Corolla and Toyota Camry: both models offer relatively low fuel consumption per kilometre, which helps protect a driver's margin on lower-fare trips, while remaining reliable and inexpensive to maintain in the Nigerian market.

Insight 6: Why Has Bolt Not Updated Its App Fare for Nigerian Drivers?

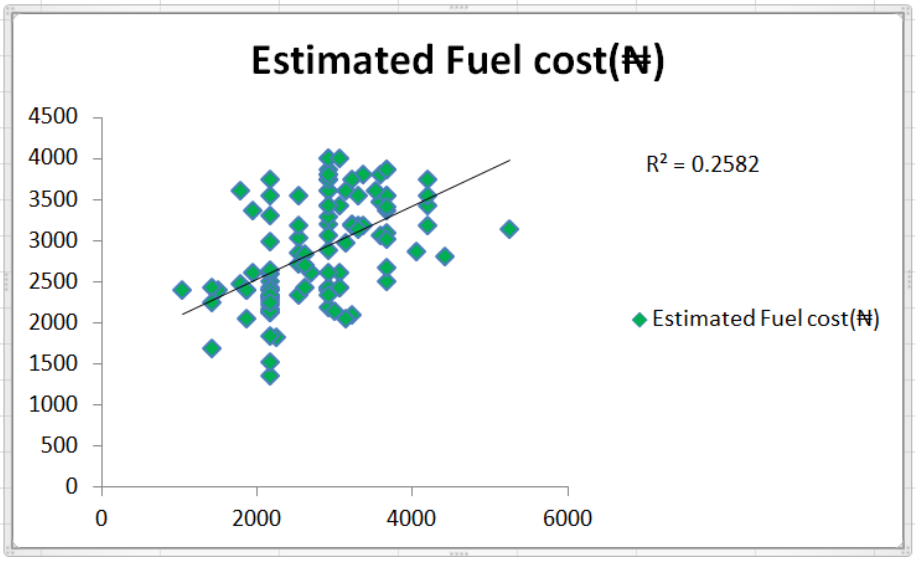


Figure 6. Estimated fuel cost plotted against driver take-home earnings, with regression trendline.

A regression analysis between estimated fuel cost and driver take-home earnings (after Bolt's 25% commission) produced an R^2 of 0.258, meaning fuel cost explains only about 26% of the variation in what a driver actually takes home. This weak relationship suggests fuel cost alone is not a strong driver of earnings in the dataset, and may help explain why Bolt has not revised its Nigerian fare structure: if Bolt's own pricing model, like the estimate used here, relies on generalised fuel efficiency assumptions rather than the real, day-to-day cost pressures drivers face — including fuel fluctuation, government policy, poor road conditions, and traffic — it may be structurally underestimating the true cost of driving in a market such as Nigeria. This remains an informed inference rather than a confirmed explanation, since Bolt's internal pricing methodology is proprietary and not available for direct verification.

Interactive Dashboard

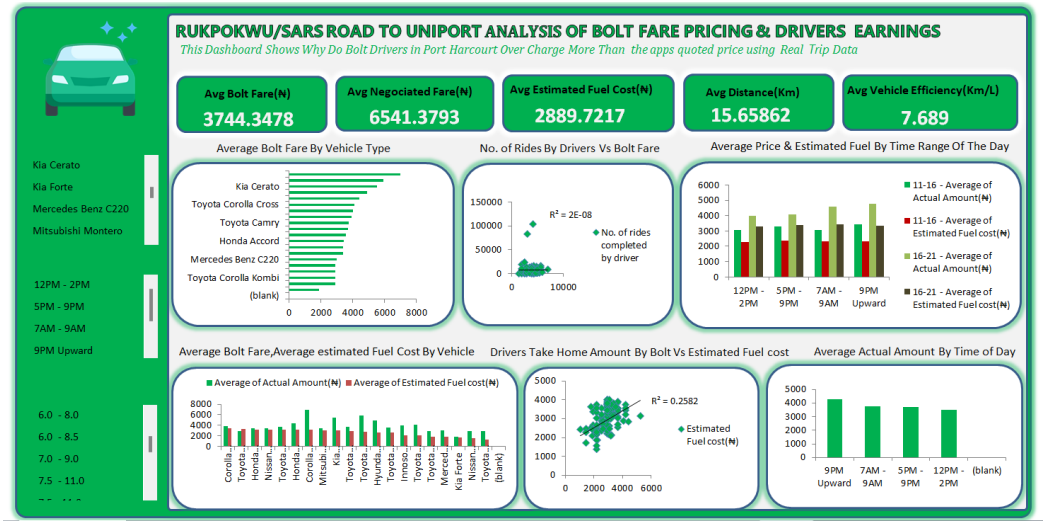


Figure 7. Full interactive Excel dashboard, filterable by time range, vehicle type, and route.

The dashboard consolidates all six insights into a single interactive view, using slicers connected to the underlying PivotTables. This allows a reader to filter the analysis by time-of-day range, vehicle type, or other variables, and immediately see how the fare, fuel cost, and driver earnings figures respond.

8. Interpretation and Business Impact

Taken together, these findings point to a pricing model that has not kept pace with the real cost of operating a ride-hailing vehicle in Nigeria. Fuel cost alone does not strongly predict what a driver takes home, which suggests that Bolt's fare algorithm — much like the estimate used in this study — may be built on generalised assumptions about fuel efficiency and running cost, rather than the specific, volatile conditions drivers face in cities like Port Harcourt. These conditions include frequent fuel price changes, government policies that push drivers towards more expensive informal fuel vendors, heavy traffic that increases real fuel consumption beyond rated efficiency, and road conditions that add wear, time, and cost to every trip.

The practical effect is that many drivers feel the app-quoted fare does not adequately cover their costs and respond by negotiating a higher amount directly with passengers. This creates friction for passengers, inconsistency in pricing across similar trips, and a risk to driver retention if the gap between quoted and actual fares continues to widen. For Bolt, closing this gap even partially could reduce the need for informal negotiation, improve the predictability of the passenger experience, and strengthen driver loyalty to the platform.

9. Recommendations

- Introduce a structured driver feedback system that captures fuel price, traffic conditions, and trip difficulty for every ride, not only the passenger's star rating for the driver.
- Review and update the in-app fare algorithm at more frequent, defined intervals (for example, every six months), particularly in markets where fuel regulation has changed, such as Nigeria following the removal of the fuel subsidy.
- Incorporate localised, real-world fuel and operating cost data into the fare model rather than relying solely on generalised or manufacturer-stated vehicle efficiency figures.

- Introduce a loyalty or reward system for drivers with a strong completed ride history to encourage retention and recognise experienced drivers, since ride count currently has no bearing on the fare offered.

10. Limitations and Next Steps

10.1 Fuel Cost as an Estimated Proxy Variable

Fuel cost was not directly recorded from drivers but estimated using publicly available fuel efficiency ratings for each vehicle model, applied under a worst-case consumption assumption. While this approach allowed for a consistent, comparable fuel cost figure across approximately 100 different vehicle models, it does not reflect actual fuel expenditure, which varies with real-world factors such as traffic congestion, driving style, vehicle maintenance condition, and daily fluctuations in fuel price. The use of a worst-case rather than an average consumption figure likely inflated fuel cost estimates uniformly across vehicles, compressing the variation that would otherwise exist between more and less efficient cars. This is a plausible explanation for why the scatter plot of negotiated fare against fuel cost showed no clear linear patterns, and why fuel cost was not a statistically significant predictor in the regression model ($p = 0.452$). Future research would benefit from driver-reported or metered fuel consumption data to validate or refine this relationship.

10.2 Fuel Cost and Driver Take-Home Earnings — A Nigerian Perspective

Regression analysis between estimated fuel cost and driver net fare (after Bolt's 25% commission) yielded an R^2 of 0.258, indicating that fuel cost explains only about 26% of the variation in what drivers actually take home per ride. This finding, while derived from a Port Harcourt-based sample, offers insight that likely extends to the broader Nigerian ride-hailing landscape, were similar structural and economic conditions shape driver operating costs across major cities.

Nigerian fuel markets are characterised by notable volatility: pump prices can shift within the same week, fuel price changes often force drivers to increase fare prices, and traffic congestion in dense urban corridors significantly increases real-world fuel consumption beyond manufacturer-rated efficiency benchmarks. Port Harcourt was used as a representative case study to examine these dynamics in practice: areas such as Rupukuw market junction and Eliozu Flyover illustrate how traffic density, road quality, and detours can drive actual fuel expenditure well above standardised estimates. While specific to Port Harcourt, these conditions are broadly characteristic of urban ride-hailing operations across Nigeria, suggesting that the pattern observed here is unlikely to be unique to this city alone.

Given this, it is plausible that ride-hailing platforms such as Bolt apply fare-pricing algorithms like the fuel cost model used in this research that rely on generalised or manufacturer-stated efficiency benchmarks rather than the volatile day-to-day operating realities drivers face across Nigerian cities. If so, the industry's broader reluctance to meaningfully adjust in-app fares may not reflect indifference to driver costs, but rather a structural limitation in how fare algorithms account for localised economic conditions such as government policies, traffic congestion, and informal fuel pricing.

This interpretation remains an informed inference rather than a confirmed explanation, as platform-specific pricing methodologies are proprietary and unavailable for direct verification. Nonetheless, using Port Harcourt as a case study allows this research to surface a broader question relevant to Nigerian ride-hailing: whether algorithm-driven

fare models adequately capture the true cost of operating in cities where fuel access and traffic conditions diverge sharply from idealised efficiency assumptions. Further studies incorporating driver-reported cost data across multiple Nigerian cities would help validate the extent to which this pattern generalises beyond Port Harcourt.

10.3 Sample Size and Scope

With 116 observations covering close to 100 different vehicle models, some vehicle categories are represented by very few data points, which limits how confidently the findings can be generalised to specific vehicle types. Similarly, since the case study is centred on a single route and city, findings should be validated against data from other Nigerian cities before being treated as universally representative.

11. Conclusion

This study set out to understand why Bolt drivers in Port Harcourt often charge more than the app's quoted fare, and why Bolt has not revised its pricing in response. Using the Rukpokwu/Sars to UNIPORT route as a case study within a wider sample of 116 real trips, the analysis found that estimated fuel cost explains only a modest share of driver take-home earnings, that revenue peaks in the evening and early-morning periods, that the Toyota Corolla and Camry dominate driver vehicle choice, and that ride count has no bearing on the fare a driver is quoted. Together, these findings suggest that Bolt's fare model may not yet reflect the full, volatile reality of driving costs in Nigeria — and that closing this gap could meaningfully benefit drivers, passengers, and the platform alike.

References

Guardian Nigeria. (2025). Bolt, Uber drivers groan over 25 per cent commission. The Guardian Nigeria. https://guardian.ng/features/executive-motoring/bolt-uber-drivers-groan-over-25-per-cent-commission/

WeeTracker. (2026). Nigeria's e-hailing drivers are taking 'offline trips' to dodge fees amid rising unease. https://weetracker.com/2026/05/14/e-hailing-offline-trips-lagos-nigeria-cash-rides/

Guardian Nigeria. (2025). Impact of fuel subsidy removal on Nigeria's economy. The Guardian Nigeria. https://guardian.ng/opinion/impact-of-fuel-subsidy-removal-on-nigerias-economy/

BusinessDay NG. (2024). How high fuel costs hindered transport sector growth in 2023. https://businessday.ng/transport/article/how-high-fuel-costs-hindered-transport-sector-growth-in-2023/